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Date and time — Overview and vocabulary

Date et l'heure — Vue d'ensemble et vocabulaire

NWIP stage

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Contents

Foreword

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This document was prepared by Technical Committee ISO/TC 154, *Processes, data elements and documents in commerce, industry and administration*.

This document is amongst a series of International Standards dealing with date and time by Technical Committee ISO/TC 154, *Processes, data elements and documents in commerce, industry and administration*, WG 5 *Representation of dates and times*.

This is the first edition of this document.

Introduction

ISO Recommendations and Standards relating to date and time concepts have been available since 1971.

This document presents terms and definitions for selected concepts relevant to date and time concepts and of their representation.

Specifically, the terminology presented in this document:

— serve as a sound basis in the understanding of date and time;

— guide new developments in the field by underpinning mutual understanding; and

— serve as a quick and handy reference for those newly inaugurated to this field.

Most terms in this document were sourced from ISO 8601-1 and ISO 8601-2.

Date and time — Overview and vocabulary

1. Scope

This document provides a concept system and a standard set of terminology related to date and time, from basic concepts to those of their representation.

The scope of the vocabulary provided in this document corresponds to that of ISO/TC 154/WG 5: Representation of dates and times.

2. Normative References

There are no normative references in this document.

3. Terms, definitions, symbols and abbreviations

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at http://www.iso.org/obp

— IEC Electropedia: available at http://www.electropedia.org

3.1. Terms and definitions

3.1.1. Basic concepts

3.1.1.1
time
mark attributed to an *instant* (3.1.1.2) by means of a specified *time scale* (3.1.1.4)

[SOURCE: IEV 113-01-12, modified — preferred term "time" placed prior to "date", replaced "date" with this preferred term "time" in Note 1, omission of Note 2, renumbering of Note 3 to Note 2, added Note 3 and Note 4 to entry]

Note 1 to entry: On a time scale consisting of successive steps, two distinct instants may be expressed by the same time (see Note 1 of the term *time scale* (3.1.1.4)).

Note 2 to entry: In common language, the term "date" is mainly used when the time scale is a calendar as a sequence of days.

Note 3 to entry: In 113-01-12 this definition corresponds with the term "date".

Note 4 to entry: The term "time" is often used in common language. However, it should only be used if the meaning is clearly visible from the context, since the term "time" is also used with other meanings.

3.1.1.2
instant
point on the *time axis* (3.1.1.3)

[SOURCE: IEV 113-01-08]

Note 1 to entry: An instantaneous event occurs at a specific instant.

3.1.1.3
time axis
mathematical representation of the succession in time of instantaneous events along a unique axis

[SOURCE: IEV 113-01-07]

Note 1 to entry: According to the special theory of relativity, the time axis depends on the choice of a spatial reference frame.

3.1.1.4
time scale
system of ordered marks which can be attributed to *instants* (3.1.1.2) on the *time axis* (3.1.1.3), one instant being chosen as the origin

[SOURCE: IEV 113-01-11, modified — references to terms defined in this document in Note 1 to entry replaced with references to their definitions in this document]

Note 1 to entry: A time scale may amongst others be chosen as:

— continuous, e.g. international atomic time (TAI) (see 713-05-18);

— continuous with discontinuities, e.g. *Coordinated Universal Time (UTC)* (3.1.1.10) due to leap sec-onds, standard time due to summer time and winter time;

— successive steps, e.g. usual calendars, where the *time axis* (3.1.1.3) is split up into a succession of consecutive time intervals and the same mark is attributed to all instants of each time interval;

— discrete, e.g. in digital techniques.

Note 2 to entry: For physical and technical applications, a time scale with quantitative marks is preferred, based on a chosen initial instant together with a unit of measurement.

Note 3 to entry: Customary time scales use various units of measurement in combination, such as second, minute, hour, or various time intervals of the calendar such as calendar day, calendar month and calendar year.

3.1.1.5
time scale unit
unit of measurement of a *time scale* (3.1.1.4)

EXAMPLE 1 calendar year, calendar month, calendar day are time scale units of the Gregorian calendar; clock hour, clock minutes, clock seconds are time scale units of the 24-hour time system clock.

3.1.1.6
clock
time scale (3.1.1.4) suited for intraday time measurements that includes the *time scale units* (3.1.1.5) of *clock second* (3.1.2.2), *clock minute* (3.1.2.4) and *clock hour* (3.1.2.6)

EXAMPLE 1 The 24-hour clock is a type of clock.

3.1.1.7
time interval
part of the *time axis* (3.1.1.3) limited by two *instants* (3.1.1.2) and, unless otherwise stated, the limiting instants themselves

[SOURCE: IEV 113-01-10, modified — merged Note 1 to entry into definition, added reference to terms defined in this document, omission of Notes 2 and 3 for clarity of this document]

3.1.1.8
recurring time interval
series of consecutive *time intervals* (3.1.1.7) of the same *duration* (3.1.1.9) or nominal duration

Note 1 to entry: If the duration of the time intervals is measured in calendar entities, the duration of each time interval depends on the calendar dates of its start and its end.

Note 2 to entry: If the time interval is repeating a set of rules, ISO 8601-2, Section 5 ("Repeat Rules for Recurring Time Intervals") applies.

3.1.1.9
duration
non-negative quantity equal to the difference between the times of the final *instant* (3.1.1.2) and the initial instant of the *time interval* (3.1.1.7)

[SOURCE: IEV 113-01-13, modified — reference to quantitative marks deleted; definition sourced from Note 1 of entry; updated references to the definitions within this document, deleted Note 1 to entry, adopted a modified Note 2 to entry and Note 3 to entry]

Note 1 to entry: The duration is one of the base quantities in the International System of Quantities (ISQ) on which the International System of Units (SI) is based. The term "time" instead of "duration" is often used in this context and also for an infinitesimal duration.

Note 2 to entry: For the term "duration", expressions such as "time" or "time interval" are often used, but the term "time" is not recommended in this sense and the term "time interval" is deprecated in this sense to avoid confusion with the concept of "time interval".

Note 3 to entry: The exact duration of a calendar year, calendar month, calendar week, calendar day, clock hour, or clock minute, depends on when they occur (for example a month can be 28, 29, 30, or 31 days, a minute can be 59, 60, or 61 seconds, etc). Therefore, exact duration can only be evaluated if the exact duration of the components are known.

3.1.1.10
Coordinated Universal Time
UTC
time scale (3.1.1.4) produced by the Bureau International des Poids et Mesures (BIPM) with the same rate as International Atomic Time (TAI), but differing from TAI only by an integral number of seconds

Note 1 to entry: TAI is a continuous time scale produced by the BIPM based on the best realizations of the SI second. TAI is a realization of Terrestrial Time (TT) with the same rate as that of TT, as defined by the International Astronomical Union Resolution B1.9 (2000).

Note 2 to entry: UTC is the time standard commonly used across the world from which local time is derived.

3.1.1.11
UTC of day
time of day (3.1.1.14) in *UTC* (3.1.1.10)

3.1.1.12
standard time
time scale (3.1.1.4) derived from *UTC* (3.1.1.10), by a time shift established in a given location by the competent authority

[SOURCE: IEV 113-01-17, modified — modified - Note 1 added to entry, added examples]

EXAMPLE 1 Central European Time (CET), Central European Summer Time (CEST), Eastern Standard Time (EST), Pacific Standard Time (PST), China Standard Time (CST), Hong Kong Time (HKT), Japanese Standard Time (JST), etc.

Note 1 to entry: This time shift may be varied in the course of a year.

3.1.1.13
local time scale
Locally-applicable *time scale* (3.1.1.4) such as *standard time* (3.1.1.12) or a non-*UTC* (3.1.1.10) based time scale

3.1.1.14
time of day
quantitative expression marking an *instant* (3.1.1.2) within a *calendar day* (3.1.2.12) by the *duration* (3.1.1.9) elapsed after the beginning of the day

[SOURCE: IEV 113-01-18, modified — "time of day" is the preferred term in this document, generalized to not require local standard time, Note 1 edited to remove examples originating from ISO 8601-1, Note 4 added]

Note 1 to entry: Usually, time of day is represented by the number of hours elapsed after beginning of day the number of minutes elapsed after the last full hour, and, if necessary, the number of seconds elapsed after the last full minute, possibly with decimal parts of a second.

Note 2 to entry: The duration used in the definition is modified for special situations (daylight saving time, *leap second* (3.1.1.21)).

Note 3 to entry: The original definition of 113-01-18 corresponds closely with that of the term "clock time", but in our definition, no *time scale* (3.1.1.4) is necessarily defined.

Note 4 to entry: The time scale used by a time of day may or may not be based on *UTC* (3.1.1.10).

3.1.1.15
local time of day
time of day (3.1.1.14) in a *local time scale* (3.1.1.13)

3.1.1.16
calendar
time scale (3.1.1.4) that uses the *time scale unit* (3.1.1.5) of *calendar day* (3.1.2.12) as its basic unit

Note 1 to entry: *calendar month* (3.1.2.20) and *calendar year* (3.1.2.22) are time scale units often included in a calendar

EXAMPLE 1 The Gregorian calendar is a type of calendar.

3.1.1.17
Gregorian calendar
calendar (3.1.1.16) that defines a calendar year (3.1.2.22) closely approximating the tropical year

Note 1 to entry: The Gregorian calendar is described in ISO 8601-1, 4.2.1.

Note 2 to entry: The Gregorian calendar is in general use.

3.1.1.18
common year
calendar year (3.1.2.22) in the *Gregorian calendar* (3.1.1.17) that has 365 *calendar days* (3.1.2.12)

3.1.1.19
leap year
calendar year (3.1.2.22) in the *Gregorian calendar* (3.1.1.17) that has 366 *calendar days* (3.1.2.12)

3.1.1.20
centennial year
calendar year (3.1.2.22) in the *Gregorian calendar* (3.1.1.17) whose year number is divisible without remainder by hundred

3.1.1.21
leap second
intentional time step of one *second* (3.1.2.1) to adjust *UTC* (3.1.1.10) to ensure appropriate agreement with UT1, a *time scale* (3.1.1.4) based on the rotation of the Earth

Note 1 to entry: See also ITU-R TF.460-6.

Note 2 to entry: An inserted second is called positive leap second and an omitted second is called negative leap second. A positive leap second is inserted after [23:59:59Z] and can be represented as [23:59:60Z]. A negative leap second is achieved by the omission of [23:59:59Z]. Insertion or omission takes place as determined by IERS, normally on 30 June or 31 December, but if necessary on 31 March or 30 September.

3.1.2. Time and Date Units

3.1.2.1
second
base unit of measurement of time in the International System of Units (SI) as defined by the International Committee of Weights and Measures (CIPM, i.e. Comité International des Poids et Mesures)

Note 1 to entry: It is the base unit for expressing duration.

Note 2 to entry: Also see ISO 80000-3.

3.1.2.2
clock second
time interval (3.1.1.7) whose duration is one *second* (3.1.2.1), which in the 24-hour *clock* (3.1.1.6) is represented by a two-digit ordinal number from [00] identifying the first second to [59] or [60] identifying the last second of a clock minute ([59] without a leap second, [60] with a leap second)

Note 1 to entry: Clock second is in common parlance often referred to as second, however in this document clock second and second have different definitions.

3.1.2.3
minute
unit of time, equal to 60 *seconds* (3.1.2.1)

Note 1 to entry: The duration of a minute is 60 seconds except if modified by the insertion or deletion of a leap second

Note 2 to entry: Also see ISO 80000-3.

3.1.2.4
clock minute
time interval (3.1.1.7) whose duration is one *minute* (3.1.2.3), which in the 24-hour clock is identified by a two-digit ordinal number from [00] identifying the first minute to [59] identifying the last minute of a clock hour

Note 1 to entry: Clock minute is in common parlance often referred to as minute, however in this document clock minute and minute have different definitions.

3.1.2.5
hour
unit of time, equal to 60 *minutes* (3.1.2.3)

Note 1 to entry: Also see ISO 80000-3.

3.1.2.6
clock hour
time interval (3.1.1.7) of duration one *hour* (3.1.2.5), which in the 24-hour clock system is identified by a two-digit ordinal number from [00] identifying the first hour through [23] identifying the last hour of a calendar day

Note 1 to entry: Clock hour is in common parlance often referred to as hour, however in this document clock hour and hour have different definitions.

3.1.2.7
date
calendar date (3.1.2.8), *ordinal date* (3.1.2.9), or *week date* (3.1.2.10)

3.1.2.8
calendar date
particular *calendar day* (3.1.2.12) represented by its *calendar year* (3.1.2.22), its *calendar month* (3.1.2.20), and its ordinal number within its calendar month

3.1.2.9
ordinal date
date (3.1.2.7) representing a particular *calendar day* (3.1.2.12) represented by its *calendar year* (3.1.2.22) and its ordinal number within its calendar year

3.1.2.10
week date
particular *calendar day* (3.1.2.12) represented by the *calendar year* (3.1.2.22) to which its *calendar week* (3.1.2.22) belongs, the ordinal number of its calendar week within that calendar year, and its ordinal number within its *calendar week* (3.1.2.17)

3.1.2.11
day
duration (3.1.1.9) of a *calendar day* (3.1.2.12)

Note 1 to entry: The term "day" applies also to the duration of any *time interval* (3.1.1.7) which starts at a certain time of day on a certain calendar day and ends at the same time of day on the next calendar day.

Note 2 to entry: Also see ISO 80000-3.

3.1.2.12
calendar day
time interval (3.1.1.7) starting at the beginning of the day and ending with the beginning of the next day (the latter being the starting instant of the next calendar day)

Note 1 to entry: Calendar day is in common parlance often referred to as day, however in this document calendar day and day have different definitions.

Note 2 to entry: The duration of a calendar day using the 24-hour clock is 24 hours; except if modified by:

— the insertion or deletion of leap seconds, by decision of the International Earth Rotation Service (IERS), or

— the insertion or deletion of other time intervals, as may be prescribed by local authorities to alter the time scale of local time.

3.1.2.13
calendar day of week
day (3.1.2.11) amongst the sequence of *week* (3.1.2.16) days, namely, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday and Sunday

Note 1 to entry: The calendar day of week is represented by a single digit ordinal number identifying a calendar day within a week, from [1] (identifying Monday) through [7] (identifying Sunday)

3.1.2.14
calendar day of month
ordinal number of a *calendar day* (3.1.2.12) within a *calendar month* (3.1.2.20)

Note 1 to entry: The calendar day of month is represented by a two-digit ordinal number identifying the calendar day within a calendar month from [01] identifying the first calendar day of a calendar month) through [28], [29], [30], or [31] (depending on the month) identifying the last day of the month.

3.1.2.15
calendar day of year
ordinal number of a *calendar day* (3.1.2.12) within a *calendar year* (3.1.2.22)

Note 1 to entry: The calendar day of year is represented by a three-digit ordinal number identifying the calendar day within a calendar year, from [001] (identifying January 1) through [365] (common year) or [366] (leap year) identifying January 31

3.1.2.16
week
duration (3.1.1.9) of a *calendar week* (3.1.2.17)

Note 1 to entry: The term "week" applies also to the duration of any time interval which starts at a certain time of day at a certain calendar day and ends at the same time of day at the same calendar day of the next calendar week.

3.1.2.17
calendar week
any seven *calendar day* (3.1.2.12) *time interval* (3.1.1.7) within a single *calendar year* (3.1.2.22) which begins on Monday and ends on Sunday

3.1.2.18
calendar week of year
ordinal number of a *calendar week* (3.1.2.17) within a *calendar year* (3.1.2.22), according to the rule that the first one is the week including the first Thursday of that year, and that the last one is the week immediately preceding the first calendar week of the next calendar year

Note 1 to entry: A calendar week of year is identified by a two-digit ordinal number from [01] for the first calendar week through [52] or [53] (depending on the number of calendar weeks in that calendar year)

3.1.2.19
month
duration (3.1.1.9) of a *calendar month* (3.1.2.20)

Note 1 to entry: The term "month" applies also to the duration of any time interval which starts at a certain time of day at a certain calendar day of the calendar month and ends at the same time of day at the same calendar day of the next calendar month, if it exists.

Note 2 to entry: In certain applications a month is considered as a duration of 30 calendar days.

3.1.2.20
calendar month
time interval (3.1.1.7) resulting from a defined division of a *calendar year* (3.1.2.22), each containing a specific number of *calendar days* (3.1.2.12)

Note 1 to entry: In the Gregorian calendar, each calendar month within its calendar year is identified by a specific name, and represented by a two-digit ordinal number from [01] for January to [12] for December.

Note 2 to entry: See ISO 8601-1, 4.2.1 for the names of the months of the calendar year in the *Gregorian calendar* (3.1.1.17), listed in their order of occurrence, for their number of days, and for the ordinal dates of the days in common and leap years.

Note 3 to entry: A calendar month is in common parlance often referred to as month, however in this document calendar month and month have different definitions.

3.1.2.21
year
duration (3.1.1.9) of a *calendar year* (3.1.2.22)

Note 1 to entry: In the *Gregorian calendar* (3.1.1.17), a year has 365 or 366 *days* (3.1.2.11). The duration is 366 days if the corresponding time interval begins February 28 or earlier in a leap year or March 2 or later in a year immediately preceding a leap year. If the interval begins February 29 (on a leap year), or March 1 of a year preceding a leap year, the end date has to be agreed on. Otherwise the duration is 365 days.

Note 2 to entry: The term "year" applies also to the duration of any time interval which starts at a certain time of day at a certain calendar date of the calendar year and ends at the same time of day at the same calendar date of the next calendar year with the exception noted in note 1.

3.1.2.22
calendar year
time interval (3.1.1.7) defined by the *calendar* (3.1.1.16) system

Note 1 to entry: The *Gregorian calendar* (3.1.1.17) defines a calendar year to be either 365 or 366 days, which begins on January 1 and ends on December 31, each Gregorian calendar year can be identified by a 4-digit ordinal number beginning with [0000] for year zero, through [9999].

Note 2 to entry: The number of digits may exceed 4 in the case of expanded representation.

Note 3 to entry: The year number may be preceded by a minus sign to indicate a year preceding year zero.

3.1.2.23
decade
time interval (3.1.1.7) unit of 10 *calendar years* (3.1.2.22), beginning with a year whose year number is divisible without remainder by ten , in the *Gregorian calendar* (3.1.1.17)

Note 1 to entry: Decade is also used to refer to an arbitrary duration of 10 years, however decade is not used as such in this document.

3.1.2.24
century
time interval (3.1.1.7) of 100 *calendar years* (3.1.2.22) duration, beginning with a *centennial year* (3.1.1.20), in the *Gregorian calendar* (3.1.1.17)

Note 1 to entry: Century is also used to refer to an arbitrary *duration* (3.1.1.9) of 100 years, however century is not used as such by this document.

EXAMPLE 1 The "19th century" covers the years 1800 through 1899, however in common parlance, it is sometimes considered to cover the years 1801 through 1900.

3.1.3. Representations and formats

3.1.3.1
date and time expression
expression indicating a *time* 3.1.1.1, *time interval* (3.1.1.7) or *recurring time interval* 3.1.1.8

EXAMPLE 1 '2018-08-01' is a date and time expression that indicates the first day of August of 2018 in the *Gregorian calendar* (3.1.1.17).

3.1.3.2
date and time representation
expression describing the format of one or more *date and time expressions* (3.1.3.1)

EXAMPLE 1 [date] is a date and time representation that can be expanded as [year][month][day], which itself can be expanded into [YYYY][MM][DD]; '20180801' is a date and time expression that conforms to this representation.

3.1.3.3
time scale component
representation of a *time scale unit* (3.1.1.5) within a *date and time expression* (3.1.3.1) or a *date and time representation* (3.1.3.2)

Note 1 to entry: A time scale component is considered of a higher order if the time scale unit it represents has a strictly larger time interval than that of another, which is considered to be a lower order component.

Note 2 to entry: Common usage of this term often omits the leading phrase "time scale", such as representing a "time scale component calendar year" by just "calendar year component". This usage is deemed accepted in this document.

EXAMPLE 1 calendar year, calendar month, calendar day, hour, minutes, seconds are time scale components of a complete representation.

3.1.3.4
basic format
date and time representation (3.1.3.2) that does not include separators between its *time scale components* (3.1.3.3)

3.1.3.5
extended format
extension of the *basic format* (3.1.3.4) that includes separators between its *time scale components* (3.1.3.3)

3.1.3.6
complete representation
date and time representation (3.1.3.2) that includes all the *time scale components* (3.1.3.3) associated with the expression

3.1.3.7
representation with reduced precision
abbreviation of a *date and time representation* (3.1.3.2) by omission of lower order *time scale components* (3.1.3.3)

3.1.3.8
decimal representation
expansion of a *date and time representation* (3.1.3.2) by addition of a decimal fraction to the lowest order *time scale component* (3.1.3.3) of the expression

3.1.3.9
expanded representation
expansion of a *date and time representation* (3.1.3.2) to allow identification of *calendar dates* (3.1.2.8) in *calendar years* (3.1.2.22) outside the range [0000] till [9999]

3.2. Symbols and abbreviations

For the purposes of this document, the following abbreviations apply.

BIPM Bureau International des Poids et Mesures / The International Bureau of Weights and Measures

CIPM Comité International des Poids et Mesures / The International Committee of Weights and Measures

TAI International Atomic Time

UTC Coordinated Universal Time

SI International System of Units

ISQ International System of Quantities

Bibliography

[1] ISO 8601-1, *Data elements and interchange formats — Information interchange — Representation of dates and times — Part 1: Basic rules*

[2] ISO 8601-2, *Data elements and interchange formats — Information interchange — Representation of dates and times — Part 2: Extensions*

[3] ISO 80000-3, *Quantities and units — Part 3: Space and time*

[4] ITU-R TF.460-6, ITU Recommendation, *Standard-frequency and time-signal emissions*

[5] IEV, IEC 60050 (all parts), *International Electrotechnical Vocabulary*, available at http://www.electropedia.org